

JDBC Integration

JDBC Integration

```
class DatabaseConnection { 6 usages  ⤴ Thaitheya  
  
private static final String URL = 1 usage  
    "jdbc:mysql://localhost:3306/student_db";  
  
private static final String USER = "root"; 1 usage  
  
private static final String PASSWORD = "root"; 1 usage  
  
public static Connection getConnection() throws SQLException { 5 usages  ⤴ T  
  
    return DriverManager.getConnection(URL, USER, PASSWORD);
```

Outputs

Created and Inserted

```
Table created.  
Inserted Successfully  
Inserted Successfully  
  
All Students  
-----  
ID : 1  
Name : John  
Department : Computer Science  
Marks : 99.0  
-----
```

Update

After Update

ID : 1

Name : John

Department : Computer Science

Marks : 99.0

ID : 3

Name : John

Department : Computer Science

Marks : 90.0

Delete

After Delete

ID : 1

Name : John

Department : Computer Science

Marks : 99.0

ID : 3

Name : John

Department : Computer Science

Marks : 90.0

ID : 4

Name : Alice

Banking System

Menu

```
===== BANK MANAGEMENT SYSTEM =====  
1. Create Savings Account  
2. Create Current Account  
3. Deposit  
4. Withdraw  
5. Transfer  
6. Check Balance  
7. Print Statement  
8. Display All Accounts  
9. Exit  
Enter your choice:
```

Create Savings Account

```
Enter your choice: 1  
Enter Account Holder Name: Thaitheyasudan  
Enter Initial Balance: 20000  
Savings Account Created Successfully.  
Account Number: ACC1001
```

Create Current Account

```
Enter your choice: 2
Enter Account Holder Name: Shiva
Enter Initial Balance: 1000000
Current Account Created Successfully.
Account Number: ACC1002
```

Deposit to the specific Account Number

```
----- BANK MANAGEMENT SYSTEM -----
1. Create Savings Account
2. Create Current Account
3. Deposit
4. Withdraw
5. Transfer
6. Check Balance
7. Print Statement
8. Display All Accounts
9. Exit
Enter your choice: 3
Enter Account Number: ACC1002
Enter Amount: 1000
```

3. Deposit

4. Withdraw

5. Transfer

6. Check Balance

7. Print Statement

8. Display All Accounts

9. Exit

Enter your choice: 6

Enter Account Number: ACC1002

Current Balance : ₹1001000.0

Withdraw Amount

- 3. Deposit
- 4. Withdraw
- 5. Transfer
- 6. Check Balance
- 7. Print Statement
- 8. Display All Accounts
- 9. Exit

Enter your choice: 4

Enter Account Number: ACC1002

Enter Amount: 30000

Transfer from one account to another

5. Transfer

6. Check Balance

7. Print Statement

8. Display All Accounts

9. Exit

Enter your choice: 5

From Account Number: ACC1002

To Account Number: ACC1001

Enter Amount: 20000

Deposit Successful.

Transfer Successful.

Balance Updated for ACC1001

6. Check Balance
7. Print Statement
8. Display All Accounts
9. Exit

Enter your choice: 6

Enter Account Number: ACC1001

Current Balance : ₹400000.0

Print Statement

7. Print Statement
8. Display All Accounts
9. Exit

Enter your choice: 7

Enter Account Number: ACC1001

Account Number : ACC1001

Account Holder : Thaitheyasudan

Balance : ₹400000.0

Transaction History

Display All Account

8. Display All Accounts

9. Exit

Enter your choice: 8

Account Number : ACC1001

Account Holder : Thaitheyasudan

Balance : ₹40000.0

Account Number : ACC1002

Account Holder : Shiva

Balance : ₹951000.0

Exit

```
9. Exit
```

```
Enter your choice: 9
```

```
Thank you for using Banking System!
```

```
Process finished with exit code 0
```

Student Management

Menu

```
===== Student Management =====  
1. Add Student  
2. View All Students  
3. Search By Id  
4. Update Student  
5. Delete Student  
6. Calculate Average  
7. Exit  
Enter your choice:
```

Add Student

```
7. EXIT
Enter your choice: 1
Enter ID: 1
Enter Name: Thaitheyasudan
Enter Age: 14
Enter Grade: B
Enter Subject: Maths, SocialScience
Students saved successfully.
Student added successfully!
```

```
Main.java Student.java StudentService.java students.txt x
1 1,Sudan,12,A,Maths
2 1,Thaitheyasudan,14,B,Maths, SocialScience
3 |
```

View All

```
6. Calculate Average
7. Exit
Enter your choice: 2
-----
ID      NAME                AGE  GRADE  SUBJECT
-----
1      Sudan                12   A      Maths
1      Thaitheyasudan       14   B      Maths, SocialScience
-----
```


Search Student

```
6. Calculate Average
7. Exit
Enter your choice: 3
Enter Student ID: 1
-----
ID      NAME                AGE  GRADE  SUBJECT
-----
1       Sudan                12   A      Maths
-----
```

Update Student

```
Enter Student ID: 1
Select field to update
1. Name
2. Age
3. Grade
4. Subject
1
Enter New Name: Siya
Students saved successfully.
Student updated successfully.
```

Enter your choice: 5

Enter Student ID: 1

```
-----  
ID      NAME                      AGE  GRADE  SUBJECT  
-----  
1       Siya                      12   A      Maths  
-----
```

Delete Student

7. Exit

Enter your choice: 5

Enter Student ID to delete: 1

Students saved successfully.

Student deleted successfully.

7. Exit

Enter your choice: 2

```
-----  
ID      NAME                      AGE  GRADE  SUBJECT  
-----  
1       Thaitheyasudan            14   B      Maths, SocialScience  
-----
```

Average GPA

```
----- Student Management -----  
1. Add Student  
2. View All Students  
3. Search By Id  
4. Update Student  
5. Delete Student  
6. Calculate Average  
7. Exit  
Enter your choice: 6  
Average GPA: 4.33
```

Exit

```
5. Delete Student
```

```
6. Calculate Average
```

```
7. Exit
```

```
Enter your choice: 7
```

```
Students saved successfully.
```

```
Thank you!
```

```
Process finished with exit code 0
```